WEIGHTED WEAK MAJORIZATION AND SHARP SCHOENBERG-TYPE INEQUALITIES

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Abstract. We introduce the notion of weighted weak majorization to study the squared moduli of the critical points of a complex polynomial whose zeros have zero centroid. The resulting estimate strengthens the classical Schoenberg inequality by extending a single quadratic trace estimate to all upper tails of the distribution function. More precisely, let P be a complex polynomial of degree n, with zeros $z_1, z_2, \dots, z_n$ and critical points $w_1, w_2, \dots, w_{n-1}$. If $\sum_{j=1}^{n} z_j = 0$, then for every increasing convex function Φ on $\mathbb{R}_+$ with $\Phi(0) = 0$, we obtain the inequality

$$\sum_{k=1}^{n-1} \Phi(|w_k|^2) \leq \frac{n-2}{n} \sum_{j=1}^{n} \Phi(|z_j|^2).$$

The coefficient is best possible uniformly over this class of convex functions. In particular, we obtain Schoenberg-type inequalities of all orders $p \geq 2$ with coefficient $(n-2)/n$, and this coefficient is sharp when n is even. Consequently, this confirms a conjecture in [TZ25]. We also prove a sharp first-order modulus-tail inequality with coefficient $\sqrt{(n-2)/n}$, together with squared and first-order directional analogues.

1. Introduction

One of the central problems in the geometry of polynomials is to understand how the zeros of a complex polynomial govern the positions and sizes of its critical points. The classical Gauss–Lucas theorem asserts that every critical point of a nonconstant complex polynomial lies in the convex hull of its zeros; see Marden [Mar66, p. 22] and Rahman–Schmeisser [RS02, Theorem 2.1.1]. A central problem in this area is Sendov's conjecture: if all zeros of a polynomial of degree $n \geq 2$ lie in the closed unit disk, then every zero is within distance one of a critical point. For historical background, see Rahman–Schmeisser [RS02, Section 7.3] or [Zha26b, Section 1]. Tao [Tao22, Theorem 1.2] proved that there exists an absolute constant n_0 such that the conjecture holds for every $n \geq n_0$. Zhang [Zha26b, Theorem 1.2] made this threshold effective by showing that one may take $n_0 = 10^{200000}$. Very recently, Mazur [Maz26] announced a computer-assisted proof of Sendov's conjecture for every degree $n \geq 2$. The exact theorem statement has been checked by a separate Lean development, while the accompanying exposition remains a preprint under review. Tao [Tao26] subsequently gave a human-readable digestion of Mazur's argument. In particular, the digested argument proves the strict interior form: if a is a zero with $|a| < 1$, then there is a critical point ζ such that $|\zeta - a| < 1$. Together with the boundary classification, this proves the Phelps–Rodriguez conjecture [PR72, Tao26]: the strict inequality holds for every zero unless $|a| = 1$ and the polynomial is a nonzero scalar multiple of $z^n - a^n$.

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The present paper studies quantitative inequalities between symmetric functionals of the zeros and the corresponding functionals of the critical points. The principal tool is majorization, and the resulting estimates are refinements of the Schoenberg inequality.

1.1. **Majorization relations between zeros and critical points.** For two vectors $x = (x_1, \dots, x_n)$ and $y = (y_1, \dots, y_n)$ in $\mathbb{R}^n_+$, let $x^\downarrow = (x^\downarrow_1, \dots, x^\downarrow_n)$ and $y^\downarrow = (y^\downarrow_1, \dots, y^\downarrow_n)$ denote their nonincreasing rearrangements. We say that x is *weakly majorized* by y, and write $x \prec_w y$, if

$$\sum_{j=1}^k x^\downarrow_j \le \sum_{j=1}^k y^\downarrow_j, \qquad 1 \le k \le n.$$

If equality also holds for $k = n$, then x is *majorized* by y, and we write $x \prec y$. Similarly, x is *weakly log-majorized* by y, written $x \prec_{w\log} y$, if

$$\prod_{j=1}^k x^\downarrow_j \le \prod_{j=1}^k y^\downarrow_j, \qquad 1 \le k \le n.$$

If equality also holds for $k = n$, we write $x \prec_{\log} y$. For the standard theory of majorization, we refer to Marshall–Olkin–Arnold [MOA11, Chapters 1–3].

Let $z_1, z_2, \dots, z_n$ be the zeros of a complex polynomial P of degree $n \ge 2$, and let $w_1, w_2, \dots, w_{n-1}$ be its critical points. All zeros and critical points are counted with multiplicity, and this convention is used throughout. A basic majorization theorem of Schmeisser [Sch03, Corollaries 3 and 4] gives

$$(|w_1|, \dots, |w_{n-1}|, 0) \prec_{w\log} (|z_1|, \dots, |z_n|).$$

In particular, weak log-majorization implies weak majorization, and therefore

$$(|w_1|, \dots, |w_{n-1}|, 0) \prec_w (|z_1|, \dots, |z_n|).$$

Consequently, for every increasing convex function $\Phi : [0, \infty) \to \mathbb{R}$ with $\Phi(0) = 0$,

$$\sum_{k=1}^{n-1} \Phi(|w_k|) \le \sum_{j=1}^n \Phi(|z_j|),$$

and in particular

$$\sum_{k=1}^{n-1} |w_k|^p \le \sum_{j=1}^n |z_j|^p, \qquad p \ge 1.$$

A stronger geometric majorization relation was obtained independently by Malamud [Mal05, Theorem 4.7] and Pereira [Per03, Theorem 5.4]. They proved that

$$(w_1, \dots, w_{n-1}, n^{-1}\sum_{j=1}^n z_j) \prec (z_1, \dots, z_n),$$

where majorization is understood in the sense of complex vectors, through the existence of a doubly stochastic matrix. In the corresponding representation, the row that produces the centroid may be taken to have all entries equal to $1/n$; hence the sums of the entries in each of the remaining columns are $(n-1)/n$. Applying convexity row by row gives Pereira's corollary [Per03, Corollary 5.5]: every convex function $\varphi : \mathbb{C} \to \mathbb{R}$ satisfies

$$\sum_{k=1}^{n-1} \varphi(w_k) \le \frac{n-1}{n} \sum_{j=1}^n \varphi(z_j).$$

Taking $\varphi(\zeta) = |\zeta|^p$, $\varphi(\zeta) = |\operatorname{Re}\zeta|^p$, and $\varphi(\zeta) = |\operatorname{Im}\zeta|^p$, respectively, gives, for $p \geq 1$,

$$\sum_{k=1}^{n-1} |w_k|^p \leq \frac{n-1}{n} \sum_{j=1}^{n} |z_j|^p,$$

$$\sum_{k=1}^{n-1} |\operatorname{Re} w_k|^p \leq \frac{n-1}{n} \sum_{j=1}^{n} |\operatorname{Re} z_j|^p,$$

$$\sum_{k=1}^{n-1} |\operatorname{Im} w_k|^p \leq \frac{n-1}{n} \sum_{j=1}^{n} |\operatorname{Im} z_j|^p.$$

However, in the study of Schoenberg-type inequalities, as discussed in Subsection 1.2, one often imposes the zero-centroid condition; in this setting, the constant in the preceding estimates is not optimal. By developing a theory of weighted weak majorization, we improve this constant under the zero-centroid assumption.

For $x, y \in \mathbb{R}_+^n$, weak majorization $x \prec_w y$ is equivalently encoded by the tail-sum inequalities

$$\sum_{i=1}^{n} (x_i - t)_+ \leq \sum_{i=1}^{n} (y_i - t)_+, \qquad t \geq 0.$$

This distribution-function viewpoint leads to the following weighted variant. Let $x \in \mathbb{R}_+^m$, $y \in \mathbb{R}_+^\ell$, and $\gamma \geq 0$. We say that x is *weakly majorized by y with weight γ*, and write $x \prec_{w,\gamma} y$, if

$$\sum_{i=1}^{m} (x_i - t)_+ \leq \gamma \sum_{j=1}^{\ell} (y_j - t)_+, \qquad t \geq 0. \tag{1.1}$$

For $\gamma = 1$ and equal lengths, this is the usual weak majorization of nonnegative vectors. For $\gamma = 0$, it simply means that $x = 0$. It is important that, even when $m = \ell$, the relation $x \prec_{w,\gamma} y$ is generally not the same as $x \prec_w \gamma y$.

Our first main result is the sharp weighted tail form of the Schoenberg inequality.

Theorem 1.1. *Let $n \geq 2$, and let P be a complex polynomial of degree n, with zeros $z_1, z_2, \ldots, z_n$ and critical points $w_1, w_2, \ldots, w_{n-1}$, all counted with multiplicity. If $\sum_{j=1}^n z_j = 0$, then*

$$(|w_1|^2, \ldots, |w_{n-1}|^2) \prec_{w, \frac{n-2}{n}} (|z_1|^2, \ldots, |z_n|^2).$$

For every $n \geq 2$, the weight $(n-2)/n$ is optimal.

Although the method behind Theorem 1.1 is inherently quadratic, the analogous first-order modulus statement also holds.

Theorem 1.2. *Let $n \geq 2$, and let P be a complex polynomial of degree n, with zeros $z_1, z_2, \ldots, z_n$ and critical points $w_1, w_2, \ldots, w_{n-1}$. If $\sum_{j=1}^n z_j = 0$, then*

$$(|w_1|, \ldots, |w_{n-1}|) \prec_{w, \sqrt{\frac{n-2}{n}}} (|z_1|, \ldots, |z_n|).$$

Equivalently, for every $t \geq 0$,

$$\sum_{k=1}^{n-1} (|w_k| - t)_+ \leq \sqrt{\frac{n-2}{n}} \sum_{j=1}^{n} (|z_j| - t)_+. \tag{1.2}$$

For every $n \geq 3$, the constant $\sqrt{(n-2)/n}$ is optimal.

The proof requires a different argument, developed in Section 5. We also establish complete directional analogues. For $\theta \in \mathbb{R}$, put

$$L_\theta(\zeta) = \mathrm{Re}(e^{-i\theta}\zeta), \qquad \zeta \in \mathbb{C}.$$

The square-level directional result is parallel to Theorem 1.1.

Theorem 1.3. *Let $n \geq 2$, and let P be a complex polynomial of degree n, with zeros $z_1, z_2, \ldots, z_n$ and critical points $w_1, w_2, \ldots, w_{n-1}$. If $\sum_{j=1}^n z_j = 0$, then, for every $\theta \in \mathbb{R}$,*

$$(L_\theta(w_1)^2, \ldots, L_\theta(w_{n-1})^2) \prec_{w, \frac{n-2}{n}} (L_\theta(z_1)^2, \ldots, L_\theta(z_n)^2).$$

The corresponding first-order estimate for every one-dimensional projection is as follows.

Theorem 1.4. *Let $n \geq 2$, and let P be a complex polynomial of degree n, with zeros $z_1, z_2, \ldots, z_n$ and critical points $w_1, w_2, \ldots, w_{n-1}$. If $\sum_{j=1}^n z_j = 0$, then, for every $\theta \in \mathbb{R}$,*

$$(|L_\theta(w_1)|, \ldots, |L_\theta(w_{n-1})|) \prec_{w, \sqrt{\frac{n-2}{n}}} (|L_\theta(z_1)|, \ldots, |L_\theta(z_n)|).$$

1.2. **Schoenberg-type inequalities.** Let $z_1, z_2, \ldots, z_n$ be the zeros of a complex polynomial P of degree $n \geq 2$, and let $w_1, w_2, \ldots, w_{n-1}$ be its critical points. In 1986, Schoenberg [Sch86, p. 8] conjectured that, under the zero-centroid condition $\sum_{j=1}^n z_j = 0$, the quadratic inequality

$$\sum_{k=1}^{n-1} |w_k|^2 \leq \frac{n-2}{n} \sum_{j=1}^{n} |z_j|^2 \tag{1.3}$$

holds, with equality if and only if all zeros are collinear. This was proved independently by Pereira [Per03] and Malamud [Mal05, Theorem 4.10]. The estimate (1.3) has inspired higher-order generalizations, now called *Schoenberg-type inequalities*; see, for instance, [BIS99, BS99, CN06, KT16, Tan25, TZ25]. As an application of (1.3), the author [Zha26a] obtained an improvement of Schmeisser's conjecture [Sch77] under the zero-centroid condition.

Recently, Tang and the author [TZ25] confirmed two fourth-order Schoenberg-type conjectures posed by Georgiev, Gómez-Serrano, Tao, and Wagner [GGTW25, pp. 32–33, Problem 6.24], giving a complete solution to the de Bruin–Sharma problem. To answer Kushel–Tyaglov's problem [KT16], they also proposed two conjectures aimed at higher orders; see [TZ25, Conjectures 1.6 and 1.7]. The first of these is the following.

Conjecture 1.5. *Let $n \geq 4$ and $p \geq 2$. Let P be a complex polynomial of degree n, with zeros $z_1, z_2, \ldots, z_n$ and critical points $w_1, w_2, \ldots, w_{n-1}$. If $\sum_{j=1}^n z_j = 0$, then*

$$\sum_{k=1}^{n-1} |w_k|^p \leq \frac{n-2}{n} \sum_{j=1}^{n} |z_j|^p.$$

Moreover, equality is attained by $P(z) = (z-1)^{n/2}(z+1)^{n/2}$ for every even $n \geq 4$.

The cases $p = 2, 4, 6$ of Conjecture 1.5 were already known; see [TZ25, Remark 1.8]. A natural interpolation argument does not yield the sharp coefficient. Indeed, if

$$V = \{z \in \mathbb{C}^n : \sum_{j=1}^{n} z_j = 0\},$$

then the complex interpolation space between $\ell^\infty(V)$ and $\ell^2(V)$ agrees with $\ell^p(V)$ only up to equivalence of norms, not isometrically in general; this point is made in [TZ25, p. 22], using Triebel's theorem [Tri78, p. 118, Theorem 1]. The interpolation argument therefore gives only the weaker coefficient

$$\left(\frac{2(n-1)}{n}\right)^{p-2} \frac{n-2}{n}, \qquad p \geq 2.$$

Theorem 1.1 gives the following Orlicz form, of which Conjecture 1.5 is an immediate special case.

Proposition 1.6. *Let $n \geq 2$, and let P be a complex polynomial of degree n, with zeros $z_1, z_2, \ldots, z_n$ and critical points $w_1, w_2, \ldots, w_{n-1}$. If $\sum_{j=1}^{n} z_j = 0$, then, for every increasing convex function $\Phi : \mathbb{R}_+ \to \mathbb{R}$ with $\Phi(0) = 0$,*

$$\sum_{k=1}^{n-1} \Phi(|w_k|^2) \leq \frac{n-2}{n} \sum_{j=1}^{n} \Phi(|z_j|^2).$$

In particular, taking $\Phi(t) = t^{p/2}$ proves Conjecture 1.5 for every $p \geq 2$.

Similarly, Theorem 1.4 has a useful modulus consequence after averaging over directions.

Proposition 1.7. *Let $n \geq 2$, and let P be a complex polynomial of degree n, with zeros $z_1, z_2, \ldots, z_n$ and critical points $w_1, w_2, \ldots, w_{n-1}$. If $\sum_{j=1}^{n} z_j = 0$, then, for every $p \geq 1$,*

$$\sum_{k=1}^{n-1} |w_k|^p \leq \sqrt{\frac{n-2}{n}} \sum_{j=1}^{n} |z_j|^p.$$

For $n \geq 3$ and $p \geq 2$, this is weaker than Proposition 1.6; therefore the new content of this estimate lies in the range $1 \leq p < 2$.

An analogous pair of estimates holds for real and imaginary parts. We begin with the Orlicz inequalities supplied by Theorem 1.3.

Proposition 1.8. *Under the hypotheses of Theorem 1.3, for every $\theta \in \mathbb{R}$ and every increasing convex function $\Phi : \mathbb{R}_+ \to \mathbb{R}$ with $\Phi(0) = 0$,*

$$\sum_{k=1}^{n-1} \Phi\big(L_\theta(w_k)^2\big) \leq \frac{n-2}{n} \sum_{j=1}^{n} \Phi\big(L_\theta(z_j)^2\big).$$

In particular, for every $p \geq 2$,

$$\sum_{k=1}^{n-1} |L_\theta(w_k)|^p \leq \frac{n-2}{n} \sum_{j=1}^{n} |L_\theta(z_j)|^p.$$

Taking $\theta = 0$ and $\theta = \pi/2$ yields the corresponding real- and imaginary-part inequalities. The two-point family $P_a(z) = z^{n-2}(z^2 - a^2)$ shows sharpness of the coefficient at the endpoint $p = 2$.

Similarly, Theorem 1.4 has a different sharp endpoint constant.

Proposition 1.9. *Under the hypotheses of Theorem 1.4, for every $\theta \in \mathbb{R}$ and every increasing convex function $\Psi : \mathbb{R}_+ \to \mathbb{R}$ with $\Psi(0) = 0$,*

$$\sum_{k=1}^{n-1} \Psi\big(|L_\theta(w_k)|\big) \leq \sqrt{\frac{n-2}{n}} \sum_{j=1}^{n} \Psi\big(|L_\theta(z_j)|\big).$$

In particular, for every $p \geq 1$,

$$\sum_{k=1}^{n-1} |L_\theta(w_k)|^p \leq \sqrt{\frac{n-2}{n}} \sum_{j=1}^{n} |L_\theta(z_j)|^p.$$

For $\theta = 0$ and $\theta = \pi/2$, this gives the real- and imaginary-part versions. The constant $\sqrt{(n-2)/n}$ is best possible at $p = 1$.

2. Tools: weighted weak majorization and the QDQ model

We begin by recording the elementary convex-function form of (1.1). This is the only lemma used below to pass from tail estimates to Orlicz inequalities.

Lemma 2.1. *Let $x \in \mathbb{R}_+^m$, $y \in \mathbb{R}_+^\ell$, and $\gamma \geq 0$. Then $x \prec_{w,\gamma} y$ if and only if, for every increasing convex function $\Phi : \mathbb{R}_+ \to \mathbb{R}$ satisfying $\Phi(0) = 0$,*

$$\sum_{i=1}^{m} \Phi(x_i) \leq \gamma \sum_{j=1}^{\ell} \Phi(y_j).$$

Proof. Assume first that $x \prec_{w,\gamma} y$, and put

$$M = \max\{x_i, y_j : 1 \leq i \leq m,\ 1 \leq j \leq \ell\}.$$

On $[0, M]$, every increasing convex Φ with $\Phi(0) = 0$ has the Stieltjes representation

$$\Phi(u) = au + \int_{(0,M]} (u - t)_+ \, d\mu(t), \qquad 0 \leq u \leq M,$$

where $a = \Phi'_+(0) \geq 0$, and μ is the positive measure induced by the nondecreasing right derivative Φ'_+. The defining tail inequalities, including the case $t = 0$, give

$$\begin{aligned}\sum_{i=1}^{m} \Phi(x_i) &= a \sum_{i=1}^{m} x_i + \int_{(0,M]} \sum_{i=1}^{m} (x_i - t)_+ \, d\mu(t) \\ &\leq \gamma a \sum_{j=1}^{\ell} y_j + \gamma \int_{(0,M]} \sum_{j=1}^{\ell} (y_j - t)_+ \, d\mu(t) \\ &= \gamma \sum_{j=1}^{\ell} \Phi(y_j).\end{aligned}$$

Conversely, taking $\Phi(u) = (u - t)_+$ for each fixed $t \geq 0$ recovers (1.1). This proves the equivalence. $\square$

The matrix model for critical points is the familiar D-companion matrix, but we shall keep it as an $n \times n$ matrix. Thus the additional zero appears explicitly and no restriction notation is needed.

Proposition 2.2. *Let $z_1, \ldots, z_n \in \mathbb{C}$, let*

$$P(\lambda) = \prod_{j=1}^{n} (\lambda - z_j), \qquad D = \operatorname{diag}(z_1, \ldots, z_n),$$

and set

$$e = n^{-1/2}\mathbf{1}, \qquad Q = I - ee^* = I - \frac{1}{n}J,$$

where J is the all-ones matrix. If $w_1, \ldots, w_{n-1}$ are the zeros of P', then

$$\operatorname{spec}(QDQ) = \{w_1, \ldots, w_{n-1}, 0\},$$

with algebraic multiplicities.

Proof. First consider DQ. For $\lambda \notin \{z_1, \dots, z_n\}$, the matrix determinant lemma gives

$$\begin{aligned}\det(\lambda I - DQ) &= \det(\lambda I - D + Dee^*) \\ &= P(\lambda)\left(1 + e^*(\lambda I - D)^{-1}De\right) \\ &= P(\lambda)\left(1 + \frac{1}{n}\sum_{j=1}^{n}\frac{z_j}{\lambda - z_j}\right) \\ &= P(\lambda)\left(\frac{\lambda}{n}\sum_{j=1}^{n}\frac{1}{\lambda - z_j}\right) = \frac{\lambda P'(\lambda)}{n}.\end{aligned}$$

Since both sides are polynomials, the identity holds for every λ. Moreover,

$$\det(\lambda I - QDQ) = \det(\lambda I - DQ^2) = \det(\lambda I - DQ) = \frac{\lambda P'(\lambda)}{n}.$$

Here the first equality follows from $\det(\lambda I - AB) = \det(\lambda I - BA)$, initially for $\lambda \neq 0$ and then for every λ by polynomial continuation, and the second uses $Q^2 = Q$. This proves the spectral identity, including algebraic multiplicities. It is the same model as in Komarova–Rivin [KR03, Lemma 5.7], retained here in the full n-dimensional QDQ form. ☐

We shall repeatedly compare positive parts of Hermitian matrices. The next lemma fixes the convention and gives the monotonicity needed below.

Lemma 2.3. *Let X and Y be Hermitian $n \times n$ matrices. If $X \le Y$, then*

$$\operatorname{tr} X_+ \le \operatorname{tr} Y_+.$$

Proof. For a Hermitian matrix T,

$$\operatorname{tr} T_+ = \max_{0 \le S \le I} \operatorname{tr}(ST).$$

Indeed, the maximum is attained by the spectral projection of T corresponding to the positive spectrum. Hence, if $X \le Y$, then for every $0 \le S \le I$ one has $\operatorname{tr}(SX) \le \operatorname{tr}(SY)$. Taking maxima gives the result. ☐

The following eigenvalue–singular-value comparison is the Weyl–Horn majorization theorem (see Bhatia [Bha97, p. 42]) in the form needed here.

Lemma 2.4. *Let $A \in M_n(\mathbb{C})$. Denote its eigenvalues and singular values, respectively, by*

$$\lambda_1(A), \dots, \lambda_n(A) \qquad \text{and} \qquad s_1(A), \dots, s_n(A).$$

Then

$$(|\lambda_1(A)|, \dots, |\lambda_n(A)|) \prec_w (s_1(A), \dots, s_n(A)).$$

Consequently, for every increasing convex function $\psi : \mathbb{R}_+ \to \mathbb{R}$,

$$\sum_{j=1}^{n} \psi(|\lambda_j(A)|) \le \sum_{j=1}^{n} \psi(s_j(A)).$$

A second comparison passes from the real parts of eigenvalues to the spectrum of the Hermitian part.

Lemma 2.5. *Let $A \in M_n(\mathbb{C})$, and fix $\theta \in \mathbb{R}$. Put*

$$R_\theta(A) = \frac{e^{-i\theta}A + e^{i\theta}A^*}{2}.$$

If $\lambda_1, \dots, \lambda_n$ are the eigenvalues of A, and $\nu_1, \dots, \nu_n$ are the eigenvalues of the Hermitian matrix $R_\theta(A)$, then

$$(\mathrm{Re}(e^{-i\theta}\lambda_1), \dots, \mathrm{Re}(e^{-i\theta}\lambda_n)) \prec (\nu_1, \dots, \nu_n).$$

Consequently, for every convex function $\varphi : \mathbb{R} \to \mathbb{R}$,

$$\sum_{j=1}^n \varphi(\mathrm{Re}(e^{-i\theta}\lambda_j)) \le \sum_{j=1}^n \varphi(\nu_j).$$

Proof. By Schur triangularization, there is a unitary U such that $U^*AU = T$ is upper triangular with diagonal entries $\lambda_1, \dots, \lambda_n$. Then

$$U^* R_\theta(A) U = \frac{e^{-i\theta}T + e^{i\theta}T^*}{2}$$

is Hermitian and has diagonal entries $\mathrm{Re}(e^{-i\theta}\lambda_1), \dots, \mathrm{Re}(e^{-i\theta}\lambda_n)$. By the Schur–Horn theorem, the diagonal vector of a Hermitian matrix is majorized by its eigenvalue vector. Karamata's inequality gives the convex-function consequence. □

3. The squared truncation estimate

The next theorem is the heart of the paper. It is a positive-part trace estimate for the squared singular values of QDQ. The additional zero singular value is present throughout and contributes nothing to the positive parts.

Theorem 3.1. *Let $n \ge 2$. Suppose that $z = (z_1, \dots, z_n) \in \mathbb{C}^n$ satisfies $\sum_{j=1}^n z_j = 0$, and put*

$$D = \mathrm{diag}(z_1, \dots, z_n), \qquad A = QDQ.$$

If $s_1(A), \dots, s_n(A)$ are the singular values of A, then, for every $t \ge 0$,

$$\sum_{i=1}^n (s_i(A)^2 - t)_+ \le \frac{n-2}{n} \sum_{j=1}^n (|z_j|^2 - t)_+. \tag{3.1}$$

Proof. Write

$$a_j = |z_j|^2, \qquad D_a = \mathrm{diag}(a_1, \dots, a_n), \qquad u = D^*e = \frac{1}{\sqrt{n}}(\overline{z_1}, \dots, \overline{z_n})^T.$$

The centroid condition says exactly that $u \perp e$, or, equivalently, $Qu = u$. Since $Q = I - ee^*$, we have

$$A^*A = QD^*QDQ = QD_aQ - uu^*.$$

Fix $t \ge 0$, and decompose

$$a_j = b_j + r_j, \qquad b_j = (a_j - t)_+, \qquad r_j = \min(a_j, t).$$

Set

$$\beta = \sum_{j=1}^n b_j, \qquad R = \sum_{j=1}^n r_j, \qquad \rho = \frac{R}{n}, \qquad D_b = \mathrm{diag}(b_1, \dots, b_n).$$

Since $D_a - tI \le D_b$, conjugation by Q gives

$$A^*A - tQ = Q(D_a - tI)Q - uu^* \le QD_bQ - uu^* =: M_t.$$

Both A^*A and Q annihilate e, and $s_n(A) = 0$. Consequently, the eigenvalues of $A^*A - tQ$ are

$$s_1(A)^2 - t, \dots, s_{n-1}(A)^2 - t, 0.$$

It follows from Lemma 2.3 that

$$\sum_{i=1}^n (s_i(A)^2 - t)_+ = \mathrm{tr}(A^*A - tQ)_+ \le \mathrm{tr}(M_t)_+.$$

It remains to estimate $\operatorname{tr}(M_t)_+$.

First,

$$\operatorname{tr}(QD_bQ) = \operatorname{tr}(D_bQ) = \frac{n-1}{n}\beta, \qquad \operatorname{tr}(uu^*) = \|u\|^2 = \frac{\beta+R}{n}.$$

Therefore

$$\operatorname{tr} M_t = \frac{n-2}{n}\beta - \rho.$$

Next, $QD_bQ \geq 0$ and uu^* has rank at most one. Thus M_t has at most one negative eigenvalue: if a negative spectral subspace had dimension at least two, it would contain a nonzero vector orthogonal to u, and on that vector the quadratic form of M_t would be nonnegative.

We now show that the possible negative eigenvalue is no smaller than $-\rho$. Equivalently, we prove

$$M_t + \rho Q \geq 0.$$

If $\rho = 0$, then $R = 0$. For $t = 0$, this is just $M_0 = A^*A \geq 0$; for $t > 0$, it forces all $a_j = 0$, and the assertion is trivial. Assume $\rho > 0$. For any $x \in \mathbb{C}^n$, put $h = Qx$. Since $u \perp e$,

$$\langle (M_t + \rho Q)x, x\rangle = \sum_{j=1}^n (b_j + \rho)|h_j|^2 - \frac{1}{n}\left|\sum_{j=1}^n z_j h_j\right|^2.$$

By the weighted Cauchy inequality,

$$\left|\sum_{j=1}^n z_j h_j\right|^2 \leq \left(\sum_{j=1}^n \frac{a_j}{b_j+\rho}\right)\left(\sum_{j=1}^n (b_j+\rho)|h_j|^2\right).$$

It is therefore enough to prove

$$\sum_{j=1}^n \frac{a_j}{b_j+\rho} \leq n.$$

For each j,

$$\frac{a_j}{b_j+\rho} = 1 + \frac{r_j - \rho}{b_j+\rho}.$$

If $b_j = 0$, the denominator in the second term is ρ. If $b_j > 0$, then $a_j > t$, so $r_j = t$; since $r_i \leq t$ for every i, we have $\rho \leq t = r_j$, and consequently

$$\frac{r_j-\rho}{b_j+\rho} \leq \frac{r_j-\rho}{\rho}.$$

Equality holds when $b_j = 0$. Summing gives

$$\sum_{j=1}^n \frac{a_j}{b_j+\rho} \leq n + \frac{1}{\rho}\sum_{j=1}^n (r_j - \rho) = n.$$

Thus $M_t + \rho Q \geq 0$.

For a Hermitian matrix T, write $T_- := (-T)_+$. The matrix M_t has at most one negative eigenvalue and none below $-\rho$. Hence $\operatorname{tr}(M_t)_- \leq \rho$, and so

$$\operatorname{tr}(M_t)_+ = \operatorname{tr} M_t + \operatorname{tr}(M_t)_- \leq \left(\frac{n-2}{n}\beta - \rho\right) + \rho = \frac{n-2}{n}\beta.$$

Since $\beta = \sum_j (|z_j|^2 - t)_+$, this is exactly (3.1). □

The next section transfers this singular-value comparison to the critical points via the spectral model of Proposition 2.2.

4. Weighted weak majorization for critical points

We now prove the principal squared-modulus theorem and its Orlicz consequences. The point is that the truncation estimate controls all tails of the singular values; Weyl–Horn then passes from singular values to eigenvalues and hence to the critical points and the additional zero.

Proof of Theorem 1.1. Let $D = \operatorname{diag}(z_1, \dots, z_n)$ and $A = QDQ$. By Proposition 2.2, the eigenvalues of A are $w_1, \dots, w_{n-1}, 0$. For fixed $t \geq 0$, the function

$$\psi_t(r) = (r^2 - t)_+, \qquad r \geq 0,$$

is increasing and convex. Lemma 2.4 gives

$$\sum_{k=1}^{n-1} (|w_k|^2 - t)_+ \leq \sum_{i=1}^{n} (s_i(A)^2 - t)_+.$$

Theorem 3.1 bounds the last sum by

$$\frac{n-2}{n} \sum_{j=1}^{n} (|z_j|^2 - t)_+.$$

This is precisely the weighted weak majorization asserted in the theorem.

To prove optimality, suppose first that $n \geq 3$, and take

$$P_a(z) = z^{n-2}(z^2 - a^2), \qquad a \neq 0.$$

Its nonzero critical points are $\pm\sqrt{(n-2)/n}\, a$. At $t = 0$, therefore,

$$\sum_{k=1}^{n-1} |w_k|^2 = 2\frac{n-2}{n}|a|^2 = \frac{n-2}{n} \sum_{j=1}^{n} |z_j|^2.$$

Thus no smaller weight can hold uniformly. For $n = 2$, the optimal weight is 0. $\square$

The Orlicz formulation now follows directly from the tail comparison.

Proof of Proposition 1.6. Apply Lemma 2.1 to Theorem 1.1. Taking $\Phi(t) = t^{p/2}$ gives the power inequality for every $p \geq 2$. $\square$

The equality family is the same one that appears in Schoenberg's quadratic inequality, and it shows that the constant cannot be improved when n is even.

Corollary 4.1. *Let $n \geq 2$, and let P be a complex polynomial of degree n, with zeros $z_1, \dots, z_n$ and critical points $w_1, \dots, w_{n-1}$. If $\sum_{j=1}^{n} z_j = 0$, then*

$$\sum_{k=1}^{n-1} |w_k|^p \leq \frac{n-2}{n} \sum_{j=1}^{n} |z_j|^p, \qquad p \geq 2.$$

If $n = 2m$, equality holds for $P(z) = (z-1)^m(z+1)^m$. Consequently, for even n, the constant $(n-2)/n$ is sharp.

Proof. The inequality is Proposition 1.6 with $\Phi(t) = t^{p/2}$. For the equality statement, the zeros are 1 and -1, each with multiplicity m, and

$$P'(z) = 2mz(z-1)^{m-1}(z+1)^{m-1}.$$

Thus the critical points are 1 and -1, each with multiplicity $m-1$, and 0 with multiplicity one. Hence

$$\sum_{k=1}^{n-1} |w_k|^p = 2(m-1) = \frac{n-2}{n}\, 2m = \frac{n-2}{n} \sum_{j=1}^{n} |z_j|^p.$$

□

The Orlicz form is often more informative than the power corollary. For example, it gives sharp estimates for every tail layer simultaneously.

Example 4.2. Under the hypotheses of Proposition 1.6, one may take $\Phi(x) = (x-\tau)_+^r$, $\tau \geq 0$, $r \geq 1$; or $\Phi(x) = e^{\lambda x} - 1$, $\lambda > 0$; or $\Phi(x) = x \log(1+x)$.

5. The modulus-tail inequality

We now turn to the proof of Theorem 1.2. The argument does not arise by interpolating the endpoint estimate. Its decisive feature is a convex reduction in the joint variables z and t. At an extreme point of the resulting convex set, at most two zeros can lie strictly outside the truncation circle; the two resulting model configurations can then be handled by singular-value calculations.

We begin by recording the matrix endpoint from which the argument starts. Let

$$e = \frac{1}{\sqrt{n}}(1, \dots, 1)^{\mathsf{T}}, \qquad Q = I - ee^*,$$

and, for $z = (z_1, \dots, z_n) \in \mathbb{C}^n$, set

$$D_z = \operatorname{diag}(z_1, \dots, z_n), \qquad A_z = QD_zQ.$$

Thus all matrices in this section act on $\mathbb{C}^n$; the additional zero singular value of A_z is retained throughout.

Lemma 5.1 ([TZ25, Theorem 4.1])**.** *If $\sum_{j=1}^n z_j = 0$, then*

$$\|A_z\|_{S_1} \leq \sqrt{\frac{n-2}{n}} \sum_{j=1}^n |z_j|. \tag{5.1}$$

The next lemma collects the three elementary spectral facts used below. We write $s_1(A) \geq \cdots \geq s_n(A) \geq 0$ for the singular values of an $n \times n$ matrix A.

Lemma 5.2. *Let $z_1, \dots, z_n \in \mathbb{C}$ satisfy $\sum_{j=1}^n z_j = 0$. Define*

$$P(\lambda) = \prod_{j=1}^n (\lambda - z_j),$$

and let $w_1, \dots, w_{n-1}$ be the zeros of P', counted with multiplicity. Let $r_1 \geq \cdots \geq r_n$ be the nonincreasing rearrangement of $|z_1|, \dots, |z_n|$. Then the following assertions hold.

(1) The eigenvalues of A_z are $w_1, \dots, w_{n-1}, 0$.

(2) One has $s_n(A_z) = 0$ and

$$\begin{aligned} s_j(A_z) &\leq r_j \quad && (1 \leq j \leq n), \\ s_j(A_z) &\geq r_{j+2} \quad && (1 \leq j \leq n-2). \end{aligned} \tag{5.2}$$

(3) One has the exact Hilbert–Schmidt identity

$$\sum_{j=1}^n s_j(A_z)^2 = \frac{n-2}{n} \sum_{j=1}^n |z_j|^2. \tag{5.3}$$

Proof. The first assertion follows directly from Proposition 2.2. Moreover, $A_z e = 0$, and hence $s_n(A_z) = 0$. Since multiplication on either side by an orthogonal projection cannot increase any singular value, $s_j(A_z) \leq s_j(D_z) = r_j$. Moreover,

$$D_z - A_z = (I-Q)D_z + QD_z(I-Q),$$

so $\operatorname{rank}(D_z - A_z) \le 2$. Using the approximation-number formula

$$s_k(T) = \inf_{\operatorname{rank} R<k} \|T-R\|,$$

we obtain, for $1 \le j \le n-2$,

$$\begin{aligned} s_{j+2}(D_z) &\le \inf_{\operatorname{rank} R<j} \big\|D_z - ((D_z - A_z) + R)\big\| \\ &= s_j(A_z). \end{aligned}$$

This proves (5.2).

Finally, cyclicity of the trace and $Q = I - ee^*$ give

$$\begin{aligned} \|A_z\|_{S_2}^2 &= \operatorname{tr}(D_z^* Q D_z Q) \\ &= \frac{n-2}{n}\sum_{j=1}^n |z_j|^2 + \frac{1}{n^2}\left|\sum_{j=1}^n z_j\right|^2. \end{aligned}$$

The last term vanishes under the centroid condition, proving (5.3). □

The truncation parameter must be incorporated into the convex structure rather than treated after the fact. For $t \ge 0$, define

$$\Phi(z,t) = \sum_{j=1}^n (s_j(A_z) - t)_+, \qquad G(z,t) = \sum_{j=1}^n (|z_j| - t)_+.$$

Lemma 5.3. *For $z, z' \in \mathbb{C}^n$ and $t, t' \ge 0$,*

$$\Phi(z+z', t+t') \le \Phi(z,t) + \Phi(z',t'), \tag{5.4}$$

and, for every $\lambda \ge 0$,

$$\Phi(\lambda z, \lambda t) = \lambda \Phi(z,t). \tag{5.5}$$

Proof. The singular-value trace inequality and a singular-value decomposition give the variational identity

$$\Phi(z,t) = \sup_{\|X\|_{S_\infty}\le 1} \left\{\operatorname{Re}\operatorname{tr}(X^*A_z) - t\|X\|_{S_1}\right\}. \tag{5.6}$$

Indeed, the expression in braces is at most $\sum_j (s_j(A_z) - t)s_j(X)$, whose maximum under $0 \le s_j(X) \le 1$ is $\sum_j (s_j(A_z)-t)_+$; equality is attained by the partial isometry supported on the singular subspaces corresponding to $s_j(A_z) > t$. Since $z \mapsto A_z$ is linear, (5.4) and (5.5) follow immediately from (5.6). □

We shall also need a one-variable fact concerning three nonnegative roots. Its role is to make the phase comparison below global, rather than merely infinitesimal.

Lemma 5.4. *Let*

$$y_1 \ge y_2 \ge y_3 \ge 0, \qquad \widetilde{y}_1 \ge \widetilde{y}_2 \ge \widetilde{y}_3 \ge 0,$$

and suppose that

$$\begin{aligned} y_1 + y_2 + y_3 &= \widetilde{y}_1 + \widetilde{y}_2 + \widetilde{y}_3, \\ y_1y_2 + y_1y_3 + y_2y_3 &= \widetilde{y}_1\widetilde{y}_2 + \widetilde{y}_1\widetilde{y}_3 + \widetilde{y}_2\widetilde{y}_3. \end{aligned}$$

If

$$\widetilde{y}_1\widetilde{y}_2\widetilde{y}_3 \le y_1y_2y_3,$$

then

$$\widetilde{y}_3 \le y_3, \qquad \widetilde{y}_2 \ge y_2, \qquad \sqrt{\widetilde{y}_1} + \sqrt{\widetilde{y}_2} \ge \sqrt{y_1} + \sqrt{y_2}. \tag{5.7}$$

Proof. Denote the common first two elementary symmetric functions by S and E, and write $x = y_3$. The other two entries satisfy

$$y_1 + y_2 = S - x, \qquad y_1 y_2 = E - Sx + x^2, \tag{5.8}$$

so the product of all three entries is

$$h(x) = x(E - Sx + x^2).$$

For every admissible smallest entry x,

$$h'(x) = E - 2Sx + 3x^2 = (y_1 - x)(y_2 - x) \geq 0.$$

Moreover,

$$S^2 - 3E = \frac{1}{2}\big((y_1 - y_2)^2 + (y_1 - y_3)^2 + (y_2 - y_3)^2\big) \geq 0.$$

Thus h' has two real roots; denote them by $\alpha \leq \beta$. Since $x \leq S/3$ and $\beta \geq S/3$, the preceding inequality forces $x \leq \alpha$, apart from the trivial case $y_1 = y_2 = y_3$. Thus every admissible smallest entry lies on the same interval on which h is increasing. The product inequality therefore implies $\widetilde{y}_3 \leq y_3$.

For fixed S and E, the smaller of the two roots in (5.8) is

$$y_2(x) = \frac{S - x - \sqrt{S^2 + 2Sx - 3x^2 - 4E}}{2}.$$

At points where the two roots are distinct,

$$y_2'(x) = -\frac{1}{2} - \frac{S - 3x}{2\sqrt{S^2 + 2Sx - 3x^2 - 4E}} \leq 0;$$

the remaining cases follow by continuity. Hence $\widetilde{y}_3 \leq y_3$ implies $\widetilde{y}_2 \geq y_2$. Finally, if $J(x) = \sqrt{y_1(x)} + \sqrt{y_2(x)}$, then

$$J(x)^2 = S - x + 2\sqrt{E - Sx + x^2},$$

and, whenever the expression is nonzero,

$$\frac{d}{dx} J(x)^2 = -1 + \frac{2x - S}{\sqrt{E - Sx + x^2}} < 0.$$

Continuity again covers the boundary cases and proves the last inequality in (5.7). □

We now turn to the proof of the theorem.

Proof of Theorem 1.2. The case $n = 2$ is immediate: the centroid condition gives the zeros a and $-a$, and the unique critical point is 0.

Suppose next that $n = 3$. Vieta's formulas give $w_1 + w_2 = \frac{2}{3}(z_1 + z_2 + z_3) = 0$, so the two critical points are w and $-w$ for some $w \in \mathbb{C}$. Put $\rho = |w|$ and $r_j = |z_j|$. By Proposition 2.2, the eigenvalues of A_z are $w, -w, 0$. Lemma 2.4, applied with $\psi(u) = u$, therefore gives $2\rho \leq \|A_z\|_{S_1}$. Hence Lemma 5.1 gives

$$2\rho \leq \frac{1}{\sqrt{3}} \sum_{j=1}^{3} r_j.$$

For $0 \leq t < \rho$, set

$$F(t) = \frac{1}{\sqrt{3}} \sum_{j=1}^{3} (r_j - t)_+ - 2(\rho - t).$$

The function F is continuous and piecewise linear. At every point of differentiability,

$$F'(t) = 2 - \frac{1}{\sqrt{3}} \#\{j : r_j > t\} \geq 2 - \sqrt{3} > 0.$$

It follows that F is nondecreasing and $F(t) \geq F(0) \geq 0$. For $t \geq \rho$ the left-hand side of (1.2) is zero. This proves the cubic case.

We may therefore assume that $n \geq 4$. Write

$$\gamma = \sqrt{\frac{n-2}{n}}, \qquad V = \left\{ z \in \mathbb{C}^n : \sum_{j=1}^{n} z_j = 0 \right\},$$

and consider the closed convex set

$$\mathcal{C} = \{(z,t) \in V \times [0,\infty) : G(z,t) \leq 1\}. \tag{5.9}$$

Its recession cone is

$$\operatorname{rec} \mathcal{C} = \{(a,s) \in V \times [0,\infty) : |a_j| \leq s \quad \text{for every } j\}. \tag{5.10}$$

Indeed, the displayed condition implies $|z_j + \lambda a_j| - (t + \lambda s) \leq |z_j| - t$ for every $\lambda \geq 0$. Conversely, applying the definition of a recession direction at $(0,0)$ gives

$$\lambda \sum_{j=1}^{n} (|a_j| - s)_+ \leq 1 \qquad (\lambda \geq 0),$$

which forces $|a_j| \leq s$ for every j. For $(a,s) \in \operatorname{rec} \mathcal{C}$,

$$\|A_a\|_{S_\infty} \leq \|D_a\|_{S_\infty} = \max_j |a_j| \leq s,$$

and hence

$$\Phi(a,s) = 0. \tag{5.11}$$

The cone in (5.10) is pointed, so the set $\mathcal{C}$ defined in (5.9) contains no line. The finite-dimensional representation theorem for closed convex sets containing no lines [Roc70, Theorem 18.5 and Corollary 18.5.2, pp. 166–167] implies that every $(z,t) \in \mathcal{C}$ can be written as

$$(z,t) = \sum_{\nu=1}^{N} \lambda_\nu (z^{(\nu)}, t_\nu) + (a,s), \qquad \sum_{\nu=1}^{N} \lambda_\nu = 1,$$

where $\lambda_\nu \geq 0$, each $(z^{(\nu)}, t_\nu)$ is an extreme point of $\mathcal{C}$, and $(a,s) \in \operatorname{rec} \mathcal{C}$. By Lemma 5.3, positive homogeneity, and (5.11),

$$\Phi(z,t) \leq \sum_{\nu=1}^{N} \lambda_\nu \Phi(z^{(\nu)}, t_\nu) + \Phi(a,s).$$

It is therefore enough to prove

$$\Phi(z,t) \leq \gamma \tag{5.12}$$

for every extreme point (z,t) of $\mathcal{C}$.

We first identify these extreme points to the extent needed. If $t = 0$, then Lemma 5.1 gives

$$\Phi(z,0) = \|A_z\|_{S_1} \leq \gamma \sum_{j=1}^{n} |z_j| = \gamma G(z,0) \leq \gamma.$$

Suppose therefore that $t > 0$. An extreme point must satisfy $G(z,t) = 1$; otherwise sufficiently small perturbations of t in either direction would remain in $\mathcal{C}$.

Set

$$O = \{j : |z_j| > t\}, \qquad B = \{j : |z_j| = t\}, \qquad I = \{j : |z_j| < t\},$$

and write $r = |O|$ and $s = |I|$. For $j \in O \cup B$, put $u_j = z_j/|z_j|$. Introduce real variables τ and δ_j for $j \in O$, and complex variables h_j for $j \in I$, and define

$$k_j = \begin{cases} (\tau+\delta_j)u_j, & j \in O, \\ \tau u_j, & j \in B, \\ h_j, & j \in I. \end{cases}$$

Impose the three real linear conditions

$$\sum_{j=1}^{n} k_j = 0, \qquad \sum_{j \in O} \delta_j = 0. \tag{5.13}$$

The variable space has real dimension $1 + r + 2s$. If $r + 2s \geq 3$, then (5.13) has a nonzero solution. For all sufficiently small $\varepsilon > 0$, both

$$(z + \varepsilon k, t + \varepsilon\tau) \quad \text{and} \quad (z - \varepsilon k, t - \varepsilon\tau)$$

then belong to $\mathcal{C}$: on O the excesses over the new threshold are $(|z_j| - t) \pm \varepsilon\delta_j$ and their sum is unchanged; the coordinates in B remain on the threshold; and those in I remain strictly below it. The chosen nonzero solution cannot satisfy $k = 0$ and $\tau = 0$, because this would force every δ_j and h_j to vanish. Thus the two perturbed points are distinct, contradicting extremality. Hence

$$r + 2s \leq 2.$$

Since $G(z,t) = 1$, one has $r \geq 1$, and therefore

$$(r,s) = (1,0) \quad \text{or} \quad (r,s) = (2,0). \tag{5.14}$$

Thus every extreme point with $t > 0$ has either one or two coordinates strictly outside the circle of radius t, while all remaining coordinates lie on that circle.

We next dispose of the one-excess model in a form that will also be used later. Assume that the coordinate moduli are

$$t + d, t, \ldots, t, \qquad d \geq 0.$$

By (5.2), the singular values have the form

$$t + L,\ \underbrace{t, \ldots, t}_{n-3 \text{ terms}}, a, 0, \qquad L \geq 0, \quad 0 \leq a \leq t.$$

Substitution into (5.3) yields

$$2tL + L^2 + a^2 = \gamma^2(2td + d^2). \tag{5.15}$$

If $L > \gamma d$, then

$$2tL + L^2 > 2t\gamma d + \gamma^2 d^2 \geq \gamma^2(2td + d^2),$$

contradicting (5.15). Consequently,

$$\Phi(z,t) = L \leq \gamma d. \tag{5.16}$$

For an extreme point of the first type in (5.14), one has $d = G(z,t) = 1$, and (5.12) follows.

Consider now an extreme point of the second type. After relabeling, its coordinate moduli are

$$t + p, t + q, t, \ldots, t, \qquad p, q > 0, \quad p + q = 1. \tag{5.17}$$

Only the first two singular values can be strictly larger than t. If only one of them is, write the singular values as

$$t + L,\ \underbrace{t, \ldots, t}_{n-3 \text{ terms}}, a, 0, \qquad 0 \leq a \leq t.$$

Equation (5.3) becomes

$$2tL + L^2 + a^2 = \gamma^2(2t + p^2 + q^2). \tag{5.18}$$

If $L > \gamma$, then, since $p^2 + q^2 \le 1$ and $\gamma \le 1$,

$$2tL + L^2 > 2t\gamma + \gamma^2 \ge \gamma^2(2t + 1) \ge \gamma^2(2t + p^2 + q^2),$$

contradicting (5.18). Hence again $\Phi(z, t) \le \gamma$.

It remains to treat the case in which both $s_1(A_z)$ and $s_2(A_z)$ exceed t. Write

$$z_j = r_j u_j, \qquad |u_j| = 1,$$

where $r_1 = t + p$, $r_2 = t + q$, and $r_j = t$ for $j \ge 3$. Let

$$U = \operatorname{diag}(u_1, \dots, u_n), \qquad R = \operatorname{diag}(r_1, \dots, r_n), \qquad f = U^* e, \qquad g = Rf.$$

The centroid condition gives

$$e^* g = \frac{1}{n} \sum_{j=1}^{n} \overline{z_j} = 0,$$

and hence $Qg = g$. If $v_j = Q\varepsilon_j$, where ε_j is the jth coordinate vector, and

$$\Delta_1 = p(2t + p), \qquad \Delta_2 = q(2t + q),$$

then, on $\mathbb{C}^n$,

$$A_z^* A_z = t^2 Q + \Delta_1 v_1 v_1^* + \Delta_2 v_2 v_2^* - gg^*. \tag{5.19}$$

Indeed, $U^* Q U = I - ff^*$ and $R^2 = t^2 I + \Delta_1 \varepsilon_1 \varepsilon_1^* + \Delta_2 \varepsilon_2 \varepsilon_2^*$. Consequently,

$$\begin{aligned} A_z^* A_z &= QRU^* QURQ \\ &= Q(R^2 - gg^*)Q, \end{aligned}$$

which gives (5.19) because $Qg = g$. Thus $A_z^* A_z - t^2 Q$ has rank at most three. In conjunction with (5.2), this gives $s_j(A_z) = t$ for $3 \le j \le n - 2$, $s_{n-1}(A_z) \le t$, and $s_n(A_z) = 0$. Relative to the baseline $t^2 Q$, the only three squared singular values that can differ from t^2 are

$$y_1 = s_1(A_z)^2, \qquad y_2 = s_2(A_z)^2, \qquad y_3 = s_{n-1}(A_z)^2,$$

where

$$y_1 > t^2, \qquad y_2 > t^2, \qquad 0 \le y_3 \le t^2. \tag{5.20}$$

Set

$$F = [v_1, v_2, g], \qquad M = \operatorname{diag}(\Delta_1, \Delta_2, -1), \qquad \Gamma = F^* F.$$

Since $QF = F$, the perturbation in (5.19) is FMF^*, and

$$\Gamma = \begin{pmatrix} \dfrac{n-1}{n} & -\dfrac{1}{n} & \dfrac{r_1 \overline{u_1}}{\sqrt{n}} \\ -\dfrac{1}{n} & \dfrac{n-1}{n} & \dfrac{r_2 \overline{u_2}}{\sqrt{n}} \\ \dfrac{r_1 u_1}{\sqrt{n}} & \dfrac{r_2 u_2}{\sqrt{n}} & \dfrac{1}{n} \sum_{j=1}^{n} r_j^2 \end{pmatrix}.$$

Put $c = \operatorname{Re}(\overline{u_1} u_2)$. Expanding the determinant gives

$$\det \Gamma = C_0 - \frac{2 r_1 r_2}{n^2} c, \tag{5.21}$$

where C_0 depends only on the radii. Sylvester's determinant identity gives

$$\det(I_n + sFMF^*) = \det(I_3 + sM\Gamma).$$

Set $\mu_i = y_i - t^2$ for $1 \le i \le 3$. These are the three eigenvalues of FMF^* that may be nonzero, with zeros included when its rank is less than three. Let e_k be their kth elementary symmetric function. The preceding identity yields

$$\det(I_3 + sM\Gamma) = 1 + se_1 + s^2e_2 + s^3e_3.$$

The first- and second-order principal minors of $M\Gamma$ depend only on the radii. Thus e_1 and e_2 are phase-independent, whereas

$$e_3 = \det(M\Gamma) = -\Delta_1\Delta_2 \det \Gamma.$$

Since $y_i = t^2 + \mu_i$, it follows that

$$\begin{aligned} S &= y_1 + y_2 + y_3 = 3t^2 + e_1, \\ E &= y_1y_2 + y_1y_3 + y_2y_3 = 3t^4 + 2t^2e_1 + e_2. \end{aligned} \tag{5.22}$$

The quantities in (5.22) are independent of all phases, while (5.21) gives

$$\begin{aligned} y_1y_2y_3 &= t^6 + t^4e_1 + t^2e_2 + e_3 \\ &= C_1 + \frac{2\Delta_1\Delta_2r_1r_2}{n^2}c, \end{aligned} \tag{5.23}$$

where $C_1 = t^6 + t^4e_1 + t^2e_2 - \Delta_1\Delta_2C_0$ depends only on the radii. Since the coefficient of c in (5.23) is positive, the product is minimized when $c = -1$. We shall verify below that a zero-sum configuration with $c = -1$ and the prescribed radii exists. Let $\widetilde{y}_1 \ge \widetilde{y}_2 \ge \widetilde{y}_3$ denote the three exceptional squared singular values for such a configuration. By Lemma 5.4,

$$\widetilde{y}_2 \ge y_2 > t^2, \qquad \sqrt{\widetilde{y}_1} + \sqrt{\widetilde{y}_2} \ge \sqrt{y_1} + \sqrt{y_2}.$$

All singular values other than the first two, the penultimate one, and the additional zero are equal to t, whereas the penultimate one does not exceed t. Hence passing to the opposite-direction configuration cannot decrease $\Phi(\,\cdot\,, t)$.

It remains only to verify that an opposite-direction zero-sum configuration with the prescribed radii exists. Assume without loss of generality that $p \ge q$, and put $d = p - q$. Since the lengths in (5.17) already occur in a zero-sum configuration, the polygon inequality gives

$$t + p \le t + q + (n-2)t, \qquad \text{hence} \qquad d \le (n-2)t.$$

For every integer $m \ge 2$, the set of sums of m unimodular complex numbers is the closed disk of radius m. For $m = 2$ this follows from $e^{i(\theta+\alpha)} + e^{i(\theta-\alpha)} = 2\cos\alpha\, e^{i\theta}$. If the assertion holds for m and $|v| \le m+1$, choose a unimodular u in the direction of v when $v \ne 0$ and arbitrarily when $v = 0$; then $|v - u| \le m$, which proves the induction step. Since $n - 2 \ge 2$, we may therefore choose unimodular numbers $\widetilde{u}_3, \dots, \widetilde{u}_n$ satisfying

$$\sum_{j=3}^{n} \widetilde{u}_j = -\frac{d}{t}\,u,$$

for a fixed unimodular u. Thus

$$\widetilde{z} = \big((t+p)u, -(t+q)u, t\widetilde{u}_3, \dots, t\widetilde{u}_n\big). \tag{5.24}$$

The vector $\widetilde{z}$ in (5.24) is zero-sum and has the same coordinate moduli as z. By (5.23), Lemma 5.4, and (5.20),

$$\Phi(z, t) \le \Phi(\widetilde{z}, t). \tag{5.25}$$

The opposite configuration can now be split into two previously controlled pieces. Define

$$a = (qu, -qu, 0, \dots, 0), \qquad b = \widetilde{z} - a.$$

Both a and b are zero-sum. Moreover,

$$\|a\|_{\ell^1} = 2q = 1 - d,$$

whereas the coordinate moduli of b are

$$t + d, t, \dots, t.$$

Applying Lemma 5.3, followed by Lemma 5.1 and (5.16), we obtain

$$\begin{aligned}\Phi(\widetilde{z}, t) &\le \Phi(a, 0) + \Phi(b, t)\\ &\le \gamma(1 - d) + \gamma d = \gamma.\end{aligned}$$

Together with (5.25), this proves (5.12) for the second type of extreme point as well.

We have therefore proved (5.12) at every extreme point of $\mathcal{C}$. The representation of $\mathcal{C}$ by its extreme points and recession cone, together with Lemma 5.3, now yields

$$G(z, t) \le 1 \quad \Longrightarrow \quad \Phi(z, t) \le \gamma.$$

Since both G and Φ are positively homogeneous in the joint variables, we conclude that

$$\Phi(z, t) \le \gamma G(z, t) \tag{5.26}$$

for all $z \in V$ and $t \ge 0$ with $G(z, t) > 0$. If $G(z, t) = 0$, then $\max_j |z_j| \le t$, whence

$$\|A_z\|_{S_\infty} \le \|D_z\|_{S_\infty} \le t.$$

Thus $\Phi(z, t) = 0$, and (5.26) remains valid. We have therefore proved, for $n \ge 4$,

$$\sum_{j=1}^{n} (s_j(A_z) - t)_+ \le \gamma \sum_{j=1}^{n} (|z_j| - t)_+.$$

It remains to pass from singular values to critical points. By Proposition 2.2, the eigenvalues of A_z are $w_1, \dots, w_{n-1}, 0$. The function

$$\psi_t(u) = (u - t)_+, \qquad u \ge 0,$$

is increasing and convex. Lemma 2.4 therefore gives

$$\sum_{j=1}^{n-1} (|w_j| - t)_+ \le \sum_{j=1}^{n} (s_j(A_z) - t)_+.$$

Combining this inequality with (5.26) proves (1.2).

Finally, let $n \ge 3$ and

$$P_a(z) = z^{n-2}(z^2 - a^2), \qquad a \ne 0.$$

Its zeros are $a, -a, 0, \dots, 0$, while

$$P_a'(z) = z^{n-3}(nz^2 - (n-2)a^2).$$

The two nonzero critical points are $\pm\sqrt{(n-2)/n}\, a$. Equality holds in (1.2) at $t = 0$, so the constant cannot be decreased. $\square$

6. Directional estimates

We next turn to real projections. The Hermitian part of $e^{-i\theta}QDQ$ is itself a real diagonal compression in the same QDQ form, and so the squared truncation theorem applies once more.

Proof of Theorem 1.3. Let $A = QDQ$. By Proposition 2.2, the eigenvalues of A are $w_1, \dots, w_{n-1}, 0$. Put

$$x_j = L_\theta(z_j) = \operatorname{Re}(e^{-i\theta} z_j), \qquad X = \operatorname{diag}(x_1, \dots, x_n).$$

The centroid condition gives $\sum_j x_j = 0$. Moreover,

$$R_\theta(A) = \frac{e^{-i\theta}A + e^{i\theta}A^*}{2} = QXQ.$$

Let $\nu_1, \dots, \nu_n$ be the eigenvalues of QXQ. By Lemma 2.5, applied to $\varphi_t(u) = (u^2 - t)_+$,

$$\sum_{k=1}^{n-1} (L_\theta(w_k)^2 - t)_+ \le \sum_{i=1}^{n} (\nu_i^2 - t)_+.$$

Since QXQ is Hermitian, its singular values are $|\nu_1|, \dots, |\nu_n|$. Applying Theorem 3.1 to the real zero-sum vector $(x_1, \dots, x_n)$ gives

$$\sum_{i=1}^{n} (\nu_i^2 - t)_+ \le \frac{n-2}{n} \sum_{j=1}^{n} (x_j^2 - t)_+.$$

This is exactly the asserted weighted weak majorization. $\square$

We next record the corresponding Orlicz consequence.

Proof of Proposition 1.8. Apply Lemma 2.1 to Theorem 1.3 to obtain the Orlicz inequality. Taking $\Phi(t) = t^{p/2}$ gives the power inequality. The choices $\theta = 0$ and $\theta = \pi/2$ give the real- and imaginary-part estimates.

For sharpness at $p = 2$, first suppose that $n \ge 3$. Let $P_a(z) = z^{n-2}(z^2 - a^2)$, where $L_\theta(a) \ne 0$. Its two nonzero critical points are $\pm\sqrt{(n-2)/n}\, a$. Therefore both sides of the directional $p = 2$ inequality are equal to $2((n-2)/n)|L_\theta(a)|^2$. When $n = 2$, the coefficient is zero and is trivially optimal. $\square$

The first-order result requires a different truncation estimate. It is purely real and depends on the geometry of the vectors $Q\varepsilon_j$, where $\varepsilon_1, \dots, \varepsilon_n$ form the standard basis of $\mathbb{C}^n$.

Lemma 6.1. *Let $x_1, \dots, x_n \in \mathbb{R}$ satisfy $\sum_{j=1}^n x_j = 0$. Put*

$$X = \operatorname{diag}(x_1, \dots, x_n), \qquad B = QXQ, \qquad \gamma_n = \sqrt{\frac{n-2}{n}}.$$

Then, for every $t \ge 0$,

$$\operatorname{tr}(|B| - tQ)_+ \le \gamma_n \sum_{j=1}^{n} (|x_j| - t)_+.$$

Proof. Let $q_j = Q\varepsilon_j$. Then

$$B = \sum_{j=1}^{n} x_j q_j q_j^*, \qquad Q = \sum_{j=1}^{n} q_j q_j^*.$$

We first prove the one-sided estimate

$$\operatorname{tr}(B - tQ)_+ \le \gamma_n \sum_{j=1}^{n} (x_j - t)_+.$$

Since $B - tQ = Q(B - tQ)Q$, the positive-trace variational formula gives

$$\operatorname{tr}(B - tQ)_+ = \max_{0\le S\le Q} \operatorname{tr}\big(S(B - tQ)\big).$$

It is therefore enough to estimate $\operatorname{tr} S(B - tQ)$ for $0 \le S \le Q$. Fix such an S, and set

$$h_j = \operatorname{tr}(Sq_jq_j^*) = \langle Sq_j, q_j\rangle \,.$$

Then $h_j \ge 0$ and $\sum_j h_j = \operatorname{tr} S$. We need the oscillation bound

$$\max_j h_j - \min_j h_j \le \gamma_n.$$

Indeed, for $j \ne k$,

$$\|q_j\|^2 = \frac{n-1}{n}, \qquad \langle q_j, q_k\rangle = -\frac{1}{n}.$$

The Hermitian matrix $q_jq_j^* - q_kq_k^*$ has trace zero, rank at most two, and Hilbert–Schmidt norm squared

$$2\left(\frac{(n-1)^2}{n^2} - \frac{1}{n^2}\right) = 2\frac{n-2}{n} = 2\gamma_n^2.$$

Its only possibly nonzero eigenvalues are therefore γ_n and $-\gamma_n$. Since $0 \le S \le Q \le I$,

$$h_j - h_k = \operatorname{tr} S(q_jq_j^* - q_kq_k^*) \le \operatorname{tr}(q_jq_j^* - q_kq_k^*)_+ = \gamma_n.$$

Interchanging j and k proves the oscillation bound.

Let $a = \min_j h_j$ and $g_j = h_j - a$. Then

$$0 \le g_j \le \gamma_n, \qquad 1 \le j \le n.$$

Since $\sum_j x_j = 0$, we have $\sum_j x_jh_j = \sum_j x_jg_j$. Also $\sum_j h_j \ge \sum_j g_j$. Hence

$$\begin{aligned}\operatorname{tr} S(B - tQ) &= \sum_{j=1}^n x_jh_j - t\sum_{j=1}^n h_j \\ &\le \sum_{j=1}^n (x_j - t)g_j \le \gamma_n \sum_{j=1}^n (x_j - t)_+.\end{aligned}$$

Taking the maximum over $0 \le S \le Q$ proves the one-sided estimate.

Applying the same estimate to $-B$ and to the zero-sum vector $(-x_1, \dots, -x_n)$ gives

$$\operatorname{tr}(-B - tQ)_+ \le \gamma_n \sum_{j=1}^n (-x_j - t)_+.$$

Since B is Hermitian and $B = QBQ$,

$$\operatorname{tr}(|B| - tQ)_+ = \operatorname{tr}(B - tQ)_+ + \operatorname{tr}(-B - tQ)_+.$$

Adding the two one-sided estimates yields the assertion. □

The directional first-order theorem is now an immediate consequence of the Hermitian-part majorization lemma.

Proof of Theorem 1.4. Let $A = QDQ$, and put

$$x_j = L_\theta(z_j), \qquad X = \operatorname{diag}(x_1, \dots, x_n), \qquad B = QXQ.$$

As above, $B = R_\theta(A)$, and $\sum_j x_j = 0$. Let $\nu_1, \dots, \nu_n$ be the eigenvalues of B. By Lemma 2.5, applied to the convex function $\varphi_t(u) = (|u| - t)_+$,

$$\sum_{k=1}^{n-1} (|L_\theta(w_k)| - t)_+ \le \sum_{i=1}^n (|\nu_i| - t)_+.$$

The right-hand side is $\operatorname{tr}(|B| - tQ)_+$, because the additional eigenvalue in the direction e is zero. Lemma 6.1 gives

$$\sum_{i=1}^{n} (|\nu_i| - t)_+ \le \sqrt{\frac{n-2}{n}} \sum_{j=1}^{n} (|x_j| - t)_+,$$

which is the desired tail inequality. □

The first-order Orlicz statement follows from the same tail principle.

Proof of Proposition 1.9. Apply Lemma 2.1 to Theorem 1.4 to obtain the first assertion. Taking $\Psi(t) = t^p$, $p \ge 1$, gives the power inequality. The real- and imaginary-part cases are obtained by setting $\theta = 0$ and $\theta = \pi/2$.

For sharpness at $p = 1$, first suppose that $n \ge 3$. Choose $a \in \mathbb{C}$ with $L_\theta(a) \ne 0$ and set $P_a(z) = z^{n-2}(z^2 - a^2)$. Its two nonzero critical points are $\pm\sqrt{(n-2)/n}\, a$, so

$$\sum_{k=1}^{n-1} |L_\theta(w_k)| = 2\sqrt{\frac{n-2}{n}} |L_\theta(a)| = \sqrt{\frac{n-2}{n}} \sum_{j=1}^{n} |L_\theta(z_j)|.$$

Thus the endpoint constant cannot be decreased. When $n = 2$, the coefficient is zero and is trivially optimal. □

Averaging the directional inequality over all directions gives the announced modulus power estimate.

Proof of Proposition 1.7. For $p \ge 1$, Proposition 1.9 gives

$$\sum_{k=1}^{n-1} |L_\theta(w_k)|^p \le \sqrt{\frac{n-2}{n}} \sum_{j=1}^{n} |L_\theta(z_j)|^p$$

for every θ. Integrating over $[0, 2\pi]$ and using

$$\int_0^{2\pi} |\operatorname{Re}(e^{-i\theta}\zeta)|^p \, d\theta = |\zeta|^p \int_0^{2\pi} |\cos\theta|^p \, d\theta$$

yields the result after cancelling the positive constant $\int_0^{2\pi} |\cos\theta|^p \, d\theta$. □

7. Sharpness and failure below order two

The coefficient $(n-2)/n$ is specific to the square-level theory. The corresponding estimate fails below order two, even for the simplest two-point family.

Proposition 7.1. *Let $n \ge 4$ and $1 \le r < 2$. There exists a complex polynomial P of degree n, with zeros $z_1, \dots, z_n$ and critical points $w_1, \dots, w_{n-1}$, all counted with multiplicity, such that $\sum_{j=1}^{n} z_j = 0$ and*

$$\sum_{k=1}^{n-1} |w_k|^r > \frac{n-2}{n} \sum_{j=1}^{n} |z_j|^r$$

holds. Moreover, for every fixed $\theta \in \mathbb{R}$, there exists such a polynomial for which

$$\sum_{k=1}^{n-1} |L_\theta(w_k)|^r > \frac{n-2}{n} \sum_{j=1}^{n} |L_\theta(z_j)|^r$$

holds.

Proof. Put $c = (n-2)/n$. For the modulus assertion, take $P_a(z) = z^{n-2}(z^2 - a^2)$, with $a \neq 0$. The zeros are a, $-a$, and 0 with multiplicity $n-2$, while the nonzero critical points are $\pm\sqrt{c}\,a$. Hence

$$\sum_{k=1}^{n-1} |w_k|^r = 2c^{r/2}|a|^r, \qquad \sum_{j=1}^{n} |z_j|^r = 2|a|^r.$$

Because $0 < c < 1$ and $r < 2$, one has $c^{r/2} > c$, so the asserted estimate with coefficient c fails.

For the directional assertion, choose a so that $L_\theta(a) \neq 0$ and use the same polynomial. Then

$$\sum_{k=1}^{n-1} |L_\theta(w_k)|^r = 2c^{r/2}|L_\theta(a)|^r, \qquad \sum_{j=1}^{n} |L_\theta(z_j)|^r = 2|L_\theta(a)|^r,$$

and the same inequality $c^{r/2} > c$ proves the failure. □

Remark 7.2. For the real part, one may take $\theta = 0$ and $a = 1$; for the imaginary part, one may take $\theta = \pi/2$ and $a = i$. The single choice $a = 1 + i$ makes both projections nonzero simultaneously.

Declaration of competing interest

The author declares no competing interests.

Data availability

No data were used in the research described in this article.

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References

[Bha97] R. Bhatia, *Matrix Analysis*, Graduate Texts in Mathematics, vol. 169, Springer, New York, 1997. doi:10.1007/978-1-4612-0653-8

[BIS99] M. G. de Bruin, K. G. Ivanov, and A. Sharma, *A conjecture of Schoenberg*, J. Inequal. Appl. **4** (1999), no. 3, 183–213. doi:10.1155/S1025583499000363

[BS99] M. G. de Bruin and A. Sharma, *On a Schoenberg-type conjecture*, J. Comput. Appl. Math. **105** (1999), 221–228. doi:10.1016/S0377-0427(99)00013-8

[CN06] W.-S. Cheung and T.-W. Ng, *A companion matrix approach to the study of zeros and critical points of a polynomial*, J. Math. Anal. Appl. **319** (2006), 690–707. doi:10.1016/j.jmaa.2005.06.071

[GGTW25] B. Georgiev, J. Gómez-Serrano, T. Tao, and A. Z. Wagner, *Mathematical exploration and discovery at scale*, arXiv:2511.02864v3 (2025). doi:10.48550/arXiv.2511.02864

[KR03] N. L. Komarova and I. Rivin, *Harmonic mean, random polynomials and stochastic matrices*, Adv. in Appl. Math. **31** (2003), no. 2, 501–526. doi:10.1016/S0196-8858(03)00023-X

[KT16] O. Kushel and M. Tyaglov, *Circulants and critical points of polynomials*, J. Math. Anal. Appl. **439** (2016), 634–650. doi:10.1016/j.jmaa.2016.03.005

[Mal05] S. M. Malamud, *Inverse spectral problem for normal matrices and the Gauss–Lucas theorem*, Trans. Amer. Math. Soc. **357** (2005), no. 10, 4043–4064. doi:10.1090/S0002-9947-04-03649-9

[Mar66] M. Marden, *Geometry of Polynomials*, 2nd ed., Mathematical Surveys, No. 3, American Mathematical Society, Providence, RI, 1966.

[MOA11] A. W. Marshall, I. Olkin, and B. C. Arnold, *Inequalities: Theory of Majorization and Its Applications*, 2nd ed., Springer Series in Statistics, Springer, New York, 2011. doi:10.1007/978-0-387-68276-1

[Maz26] L. Mazur, *A computer-assisted proof of Sendov's conjecture*, public version of August 5, 2026, ProofAtlas, 19 pp., `https://www.proofatlas.ai/formalizations/sendov-conjecture/`.

[Per03] R. Pereira, *Differentiators and the geometry of polynomials*, J. Math. Anal. Appl. **285** (2003), no. 1, 336–348. doi:10.1016/S0022-247X(03)00465-7

[PR72] D. Phelps and R. S. Rodriguez, *Some properties of extremal polynomials for the Ilieff conjecture*, Kodai Math. Sem. Rep. **24** (1972), no. 2, 172–175. doi:10.2996/kmj/1138846519

[RS02] Q. I. Rahman and G. Schmeisser, *Analytic Theory of Polynomials*, London Mathematical Society Monographs, New Series, vol. 26, Oxford University Press, Oxford, 2002. doi:10.1093/oso/9780198534938.001.0001

[Roc70] R. T. Rockafellar, *Convex analysis*, Princeton Math. Ser., No. 28, Princeton Univ. Press, Princeton, NJ, 1970, doi:10.1515/9781400873173

[Sch03] G. Schmeisser, *Majorization of the critical points of a polynomial by its zeros*, Comput. Methods Funct. Theory **3** (2004), no. 1, 95–103. doi:10.1007/BF03321027

[Sch77] G. Schmeisser, *On Ilieff's conjecture*, Math. Z. **156** (1977), 165–173. doi:10.1007/BF01178761

[Sch86] I. J. Schoenberg, *A conjectured analogue of Rolle's theorem for polynomials with real or complex coefficients*, Amer. Math. Monthly **93** (1986), no. 1, 8–13. doi:10.1080/00029890.1986.11971734

[Tan25] Q. Tang, *Schoenberg type inequalities*, arXiv:2504.09837v2 (2025). doi:10.48550/arXiv.2504.09837

[TZ25] Q. Tang and T. Zhang, *Sharp Schoenberg type inequalities and the de Bruin–Sharma problem*, arXiv:2508.10341v3 (2025), `https://arxiv.org/abs/2508.10341v3`.

[Tao22] T. Tao, *Sendov's conjecture for sufficiently high degree polynomials*, Acta Math. **229** (2022), no. 2, 347–392. doi:10.4310/ACTA.2022.v229.n2.a3

[Tao26] T. Tao, *A digestion of the proof of Sendov's conjecture*, What's New, August 12, 2026, `https://terrytao.wordpress.com/2026/08/12/a-digestion-of-the-proof-of-sendovs-conjecture/`.

[Tri78] H. Triebel, *Interpolation Theory, Function Spaces, Differential Operators*, North-Holland Mathematical Library, vol. 18, North-Holland, Amsterdam, 1978.

[Zha26a] T. Zhang, *A refinement of Pawlowski's result*, Proc. Amer. Math. Soc. **154** (2026), no. 2, 775–782. doi:10.1090/proc/17471

[Zha26b] T. Zhang, *Sendov's conjecture holds for every degree $n \geq 10^{200000}$*, preprint, 2026, 29 pp., `https://zhangteng2000.github.io/files/sendov_conjecture_large_n.pdf`.

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